

HAKADAL, Norway

WMO: 014880

Lat: 60.1172N

Lon: 10.8289E

Elev: 558

StdP: 14.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	2.0	-6.5	4.0	-1.7	-1.6	5.1	3.1	19.2	36.5	15.9	35.2	2.1	290	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	80.6	63.4	77.1	62.3	73.6	60.5	66.4	74.5	64.6	73.3	62.6	70.7	5.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	90.8	69.0	61.2	82.3	66.9	59.4	77.1	65.3	31.3	74.3	29.9	73.3	28.5	70.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.3	12.4	10.0	DB	-7.0	84.4	5.5	4.4	-11.0	87.6	-14.2	90.2	-17.3	92.7	-21.3	95.9	(4)
			WB	-7.4	67.9	5.5	3.3	-11.4	70.3	-14.6	72.3	-17.7	74.2	-21.8	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	22.4	24.7	32.3	40.7	50.6	57.7	62.7	60.2	52.2	41.9	32.8	24.4
	DBStd	15.76	9.62	9.46	7.05	5.65	7.01	4.94	5.06	5.63	5.28	6.27	7.31	10.32
	HDD50	4089	854	709	550	286	80	3	0	0	37	261	516	793
	HDD65	8473	1319	1129	1015	730	448	226	107	174	385	716	966	1258
	CDD50	1161	0	0	0	6	98	234	395	316	102	10	0	0
	CDD65	70	0	0	0	0	1	7	37	25	0	0	0	0
	CDH74	679	0	0	0	0	40	119	378	141	1	0	0	0
	CDH80	108	0	0	0	0	3	18	74	13	0	0	0	0
Wind	WSAvg	3.6	3.0	3.3	3.9	4.2	4.1	4.1	3.7	3.5	3.6	3.4	2.9	2.9
Precipitation	PrecAvg	34.4	2.9	1.9	1.6	2.2	2.4	2.9	3.5	3.6	3.1	3.9	4.0	2.4
	PrecMax	50.1	6.4	3.7	5.0	4.3	4.1	6.5	7.0	7.4	6.1	9.8	12.9	3.9
	PrecMin	27.3	0.6	0.7	0.0	0.0	0.2	0.3	0.5	0.9	0.9	0.7	0.9	0.4
	PrecStd	5.2	1.5	0.9	1.3	1.1	1.3	1.7	1.5	1.6	1.1	2.0	2.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	46.7	56.6	66.9	78.8	82.4	86.0	81.9	70.5	59.6	52.3	46.4
		MCWB	40.2	39.8	45.8	51.7	58.5	62.3	64.7	65.9	59.4	52.9	49.1	43.2
	2%	DB	39.2	42.6	50.6	60.4	73.4	77.4	82.0	78.0	67.0	56.9	48.1	43.1
		MCWB	37.5	37.7	41.3	48.4	57.0	59.7	63.7	64.3	57.1	52.4	46.1	41.3
	5%	DB	37.0	39.6	45.9	55.5	69.2	72.8	78.5	74.6	64.2	55.0	45.5	40.5
		MCWB	35.3	36.8	38.1	45.6	55.2	57.8	63.0	62.8	55.2	51.3	43.9	38.6
	10%	DB	34.4	37.2	42.5	51.8	64.3	69.0	74.8	71.1	61.5	52.7	43.1	37.2
		MCWB	33.1	35.1	36.8	43.3	52.8	55.6	61.7	60.7	55.2	49.1	41.1	35.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	42.3	46.2	53.5	61.9	64.2	68.4	68.9	62.3	56.4	50.0	44.2
		MCDB	42.0	45.4	55.5	63.2	71.2	74.8	76.6	75.2	67.7	58.3	51.5	45.4
	2%	WB	37.7	39.2	42.3	49.9	59.0	61.8	66.4	67.0	59.7	53.9	46.2	41.6
		MCDB	38.6	40.8	47.9	57.7	68.8	72.2	75.8	73.8	63.2	55.6	47.3	43.0
	5%	WB	35.5	37.2	39.6	47.3	56.6	60.1	64.8	65.0	57.9	52.1	44.3	38.6
		MCDB	36.2	39.1	43.8	53.3	65.6	69.6	74.7	72.1	62.2	54.3	45.5	39.9
	10%	WB	33.3	35.0	37.5	44.8	54.5	58.1	63.1	62.7	56.0	49.4	41.5	35.6
		MCDB	34.1	36.8	41.5	50.8	63.0	66.1	72.3	69.2	60.3	51.8	42.7	36.6
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.5	16.0	18.1	19.2	18.2	19.1	17.9	16.0	12.9	8.5	9.9
		MCDBR	10.2	13.4	21.1	25.5	27.8	25.9	25.9	20.6	19.9	14.3	12.4	12.3
		MCWBR	8.7	10.2	12.9	14.4	13.5	11.7	11.0	9.3	11.7	10.4	10.8	11.3
	5% WB	MCDBR	9.1	11.9	18.6	22.1	22.4	20.9	21.5	17.8	15.3	12.4	11.4	11.6
		MCWBR	8.3	9.8	12.3	13.2	11.6	10.7	10.2	9.1	10.2	10.3	10.4	10.8
Clear-Sky Solar Irradiance	taub	0.255	0.297	0.325	0.369	0.366	0.355	0.376	0.369	0.335	0.321	0.275	0.227	
	taud	2.194	2.272	2.313	2.305	2.397	2.448	2.409	2.425	2.506	2.453	2.279	2.105	
	Ebn at Noon	167	223	255	262	273	278	268	261	252	216	159	128	
	Edh at Noon	13	21	28	34	33	32	33	30	23	18	12	9	
All-Sky Solar Radiation	RadAvg	89	285	752	1193	1539	1761	1575	1238	819	362	119	58	
	RadStd	16	52	80	159	158	135	206	102	95	70	33	5	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (56 neighbors)	N/A	N/A	+2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page